XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

COM111 Electrical Engineering

Course Overview

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Consultation Hour: by appointment

Programme Educational Objective (PEO)

PEO1	**Professionalism** Graduates will be capable of using problem-solving techniques and critical analysis to establish themselves as practicing professionals and demonstrate leadership skills and entrepreneurial spirit in the field of Electrical and Electronics Engineering or related fields.
PEO2	**Continuous Personal Development** Graduates will engage in lifelong pursuit of knowledge and interdisciplinary learning appropriate for industrial and academic careers.
PEO3	**Societal Engagement** Graduates will contribute to sustainable development and the well-being of society.

Course Learning Outcome (CLO) and Programme Learning Outcome (PLO)

CLO1	Demonstrate the understanding of the circuit elements in the electrical system.
CLO2	Apply various circuit analysis for solving electrical engineering problems.
CLO3	Solve scientifically and critically for practical problems of electrical engineering.

PLO1	Engineering Knowledge - Apply knowledge of mathematics, science, electrical & electronic engineering fundamentals and engineering specialisation principles to define and utilize engineering procedures, processes, systems and methodologies
PLO2	Problem Analysis - Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences
PLO5	Modern Tool Usage - Create, select, and apply appropriate IT tools to model and simulate complex engineering problems by having a full understanding of their limitations.

Course Assessment

Assessment	Detail	(%)
Continuous Assessment	Assignment/Demonstration	40
	Quiz	20
Final Examination	Final exam (2 hours) Answer all questions	40
	Total	100

NO CHEATING POLICY

— Any academic misconduct is strictly prohibited. Strictly no copying allowed for assignments, quizzes and final exam.

—Discussion among students is encouraged for the assignments, but the final submission must be your own work.

Grading

Exam Questions, Marking Scheme & Invigilation	
Special Requirements (For EAC approved Engineering Programmes)	• Multiple Choice Questions (MCQ) are not encouraged in the final exams. • **Students shall pass the final examination to pass the course, the threshold marks for final examination is 40.**

Marks	Grade	Grade Points	Description
85 – 100	A	4.00	Distinction
80 – 84	A-	3.70	Distinction
75 – 79	B+	3.30	Good
70 – 74	B	3.00	Good
65 – 69	B-	2.70	Good
60 – 64	C+	2.30	Pass
55 – 59	C	2.00	Pass
50 - 54	C-	1.70	Conditional Pass
45 – 49	D+	1.30	Fail
40 - 44	D	1.00	Fail
≤ 39	F	0.00	Fail

Grading scales and grades applicable to ALL intakes effective from Academic session 2017/09

Course Plan

Chapter 1 – Introduction to Electrical Engineering

- Subdivisions of electrical engineering
- Electrical circuit, current, and voltage
- Power and energy
- Kirchhoff’s current law
- Series circuit
- Kirchhoff’s voltage law
- KVL and conservation of energy
- Parallel circuit
- Ideal circuit elements
  - Resistors
  - Conductors
  - Independent sources
  - Dependent sources
- Circuit diagram analysis

Course Plan

Chapter 2 – Resistive Circuits

- Series resistances
- Parallel resistances
- Circuit analysis using series/parallel equivalents
- Voltage/current divider rule
- Node-voltage analysis (Nodal analysis)
- Supernode
- Mesh-current analysis
- Supermesh

Course Plan

Chapter 3 – Circuit Analysis Theorems and Techniques

- Thévenin equivalent circuit
- Norton equivalent circuit
- Source transformations
- Maximum power transfer
- Superposition principle
- Wheatstone bridge

Course Plan

Chapter 4 – Capacitor and Inductor

- Ideal circuit elements: Capacitors
- Ideal circuit elements: Inductors
- Transients
- Transient response of first-order RC/RL circuits
- Transient response of second-order RLC circuits
- Second order zero-input response
- Standard form of zero-input DE
- Damping types
- Second order zero-state response

Course Plan

Chapter 5 – Steady State Sinusoidal Analysis

- Sinusoidal currents and voltages
- Phasors
- Complex impedances
- Circuit analysis with phasors and complex impedances
- Power in AC circuits
- Thévenin and Norton equivalent circuits
- Balanced three-phase circuits
- Three-phase power system

At the end of the course

You should be able:

- Design and analyze electrical circuits.
- Solve real-world engineering problems.
- Build a strong foundation for advanced studies or careers in electrical engineering.

How to learn Electrical Engineering effectively

- Read the lecture slides and textbook prior to the class.
- Do not miss any class, pay attention and participate in the class.
- Try the tutorials and examples from textbook frequently, practice makes perfect.

Attendance

- Attendance is required by the Ministry of Higher Education Malaysia.
- Sign the attendance sheet during class.
- If you cannot make it to class, sick leaves and other types of leaves should be applied 3 days prior/after, else you will be marked as absent.
- For two consecutive absences, you will be reported to Academic Affairs.

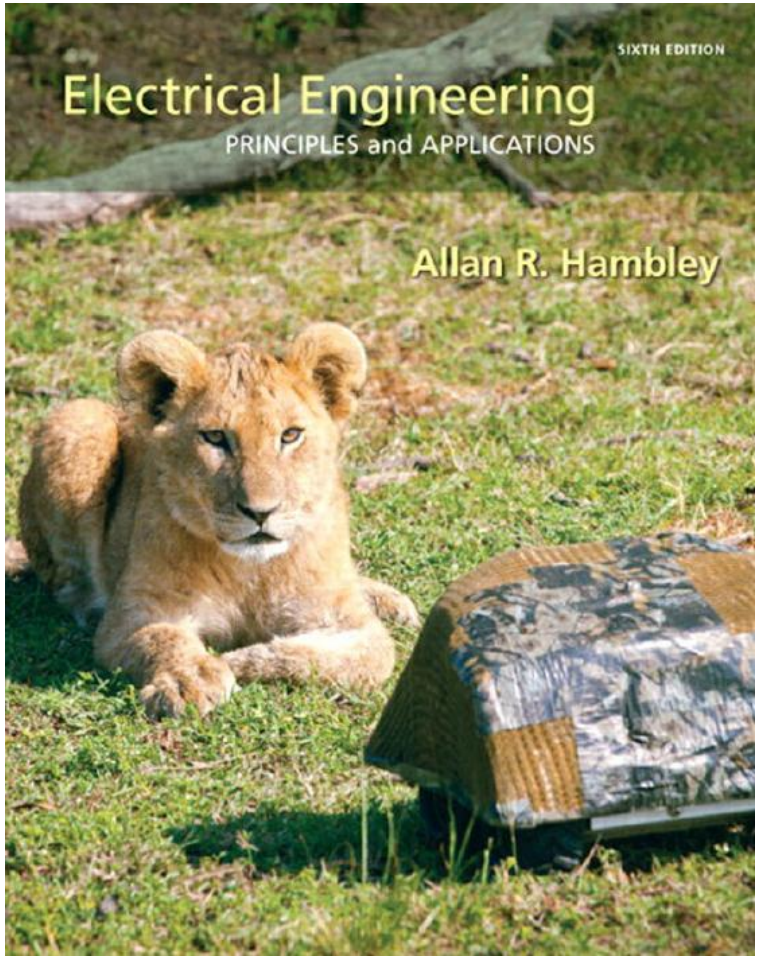
Course communication

- Email or Teams message
  - Requests for extensions, questions about course policies
  - Grading inquiries, requests for help with assignments
  - Sick leaves and other types of leaves
- Face-to-face
  - Class breaks
  - Email/message me to make an appointment for consultations

Textbook used in this course

Allan R Hambley, **Electrical Engineering: Principles and Applications, International Edition**, Pearson, 2018.

- an excellent book and easy to understand
- many examples for course study



Further reading

- Giorgio Rizzoni, **Principles and Applications of Electrical Engineering**, Mc Graw Hill, 2022